QUARTERLY OPERATIONS REVIEW

Safety, Equipment Health & Supply Chain Performance

Presented to the Senior Management Team | David Kimathi

Nairobi Depot's equipment and safety issues cost an estimated $15.8M in lost output this year — root-caused, quantified, and now moving from diagnosis to a funded early-warning pilot. Supply chain forecasting is accurate enough for planning (4.4% error) and a safety-stock policy already shows measurably fewer stockouts in simulation.

$15.8M
Value of NBI-P03's
lost throughput (illustrative pricing)

93.5%
of NBI-P03 shifts
flagged missed-maintenance

4.4%
Demand forecast
error (MAPE)

1.7%→0.6%
Simulated stockout rate,
without vs. with safety stock

Ask of this room: approve a targeted PdM sensor pilot on NBI-P03 and adopt the proposed safety-stock policy depot-wide.

Recap from Week 6

- One pump (NBI-P03) drives the entire Nairobi throughput shortfall — statistically confirmed (Welch's t-test, $p < 0.001$; large effect size).
- A prototype early-warning model catches ~90% of bad shifts (sensitivity), but is not yet a true leading indicator — it confirms problems slightly early, doesn't predict them days ahead.
- Root gap: no real vibration/pressure sensor on the equipment — only throughput and voltage proxies are available today.

SLIDE 4 · SAFETY & EQUIPMENT HEALTH (2/2)

40.6%

throughput loss per affected shift

226,000+

barrels lost annually — attributable entirely to this one machine

Recommended next step: retrofit real vibration sensors → convert this from reactive detection to true 3+ day early warning.

Recap from Week 7

- 22 slip/trip incidents at Nairobi Depot this year — 16 of 22 (73%) on the night shift, concentrated at the transfer walkway.
- A same-day safety alert was issued to night-shift crews with specific action steps; a lighting/surface inspection is underway.
- The live HSE dashboard (built this quarter) now flags critical incidents automatically and lets any site leader filter by site, date, and incident type in real time.

SLIDE 5 · SUPPLY CHAIN PERFORMANCE (1/3)

73%

of Nairobi slip/trip incidents occur on night shift

Status: interim signage up; formal lighting inspection scheduled this week.

New this quarter (Week 8)

4.4%	48.7 bbl	45 days
Forecast error (MAPE), 45-day holdout	Daily forecast uncertainty (RMSE)	Forecast horizon (within 30–60 day target)

The model captures weekly demand patterns (weekend dip), the mid-year seasonal peak, Kenyan public holidays, and known promotional periods — validated against 45 days of held-out actual demand it never saw during training.

Built with Prophet, incorporating public holidays and promotions as external regressors.

Simulated on 180 days of actual demand, same conditions, two policies compared:

WITHOUT Safety Stock

1.7%

stockout days
(3 of 180)

Reorder point: 4,920 barrels

WITH Safety Stock (95% target)

0.6%

stockout days
(1 of 180)

Reorder point: 5,099 barrels (+179 bbl safety stock)

SLIDE 7 · SUPPLY CHAIN PERFORMANCE (3/3)

Minimum-cost weekly allocation from the central terminal to all four depots (linear programming):

Depot	Weekly Volume
Nairobi	7,368 barrels
Mombasa	7,176 barrels
Kisumu	6,866 barrels
Eldoret	6,700 barrels

$24,978

total weekly shipping cost
at minimum-cost allocation

All four depots' demand is met at full capacity utilization — the terminal has almost no spare capacity today, which is itself a planning signal worth flagging.

SLIDE 8 · STRATEGIC RECOMMENDATIONS

1

Pilot vibration sensors on NBI-P03

Converts a reactive detector into a true 3+ day early-warning system.

Est. cost: low five figures | Value at risk: $15.8M/yr in lost throughput

2

Adopt the 95% safety-stock policy depot-wide

Simulation shows stockout days drop ~3x with a modest inventory increase.

Est. cost: ~179 bbl extra carrying cost/depot | Prevents lost-sale stockout events

3 Review terminal capacity headroom

Current allocation uses nearly 100% of terminal capacity — no buffer for demand spikes or a depot outage.

Est. cost: TBD (facility-dependent) | Reduces risk of a network-wide shortfall

Q&A

Anticipated CFO question:

“Why do we need so much safety stock?”

179 barrels of safety stock is not padding — it's sized directly from our actual forecast error (48.7 barrels/day) at a 95% service level, the industry-standard balance between holding cost and stockout risk. In simulation, it cut stockout days roughly 3x on the exact same demand history. The alternative isn't zero cost — it's paying for missed sales and rush orders instead of a small, quantified inventory buffer.

Thank you.